Continuous-Variable Quantum Key Distribution (CV-QKD)

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1. Why CV-QKD?

Gaussian continuous-variable (CV) Quantum key distribution (QKD) enables secure communication by leveraging the laws of quantum mechanics. Continuous-variable QKD (CV-QKD) uses coherent states and homodyne detection, making it compatible with standard telecom components – 1550 nm wavelength.

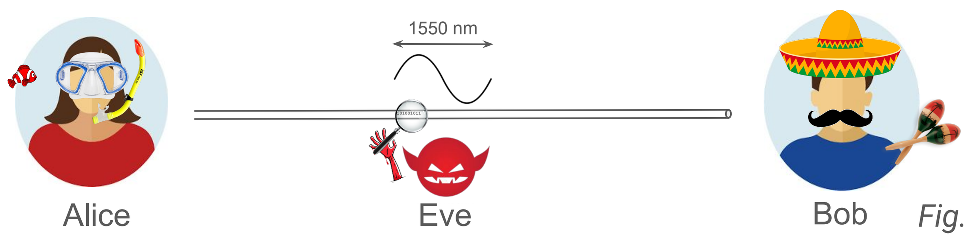

Fig. 1: Alice, Bob and Eve

2. How does it work?

Alice sends light pulses with specific quadrature values, and Bob measures these quadratures. If an eavesdropper (Eve) tries to intercept the transmission, the quantum state sent by Alice will be disturbed. This disturbance can be detected, revealing the presence of an intrusion.

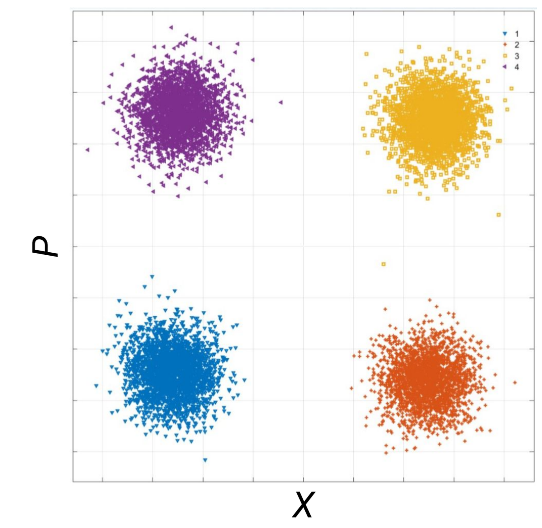

Fig. 2: Encoding information in the amplitude-phase space.

3. Experimental setup

The experimental setup consists of a tunable laser (1550 nm) which is split up into a signal and local oscillator (LO) with a 90:10 coupler. Using, amplitude and phase modulators (AM/PM), Alice can encode and send a message to Bob. Using a phase shifter (PS), Bob locks the phase between signal and LO and performs a homodyne detection (HD) of X or P. To avoid saturation of the photodiodes in the homodyne detector, the LO is attenuated with a variable attenuator (VAAT). The polarization controller is used to match polarization of signal and LO, tuning the interference between them.

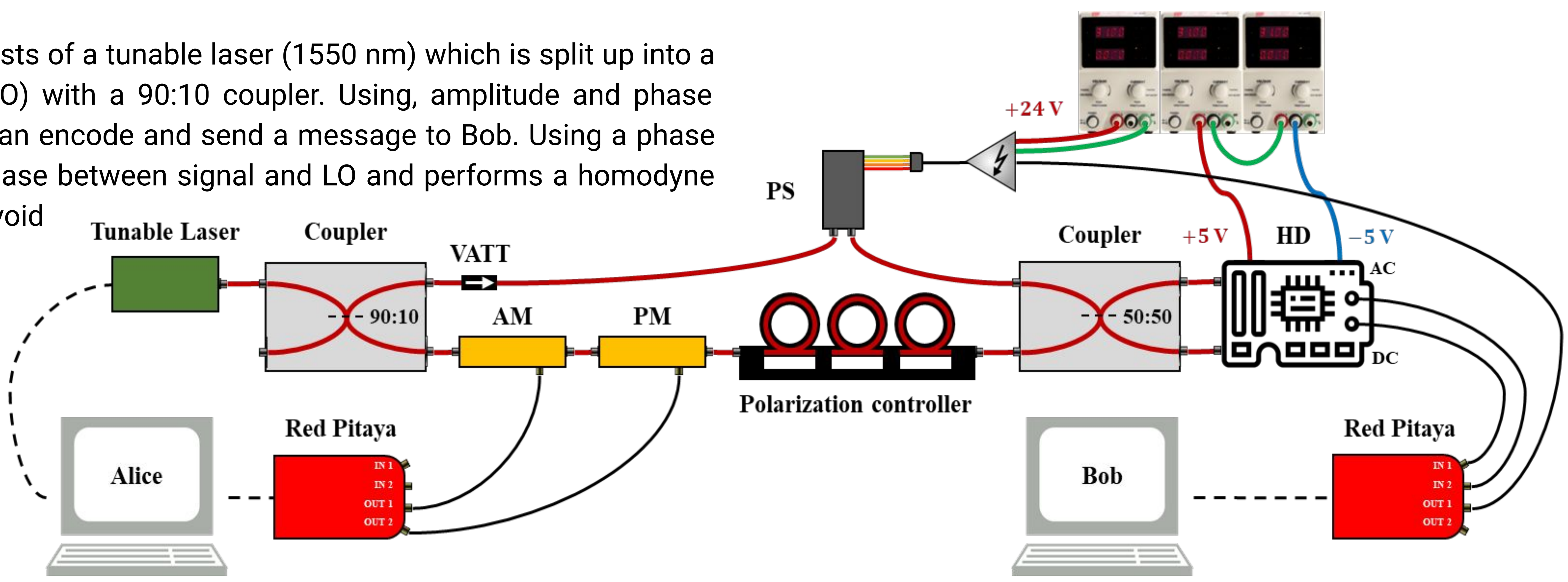

Fig. 3: Principle sketch of the experimental setup used for CV-QKD. Alice encodes key in amplitude/phase and sends to Bob who decodes and obtains the key.

4. Characterization

i. Homodyne Detector

The homodyne detector was balanced and the photodiodes characterized by measuring the DC voltage for different laser powers. Estimated efficiencies are 0.761 and 0.727.

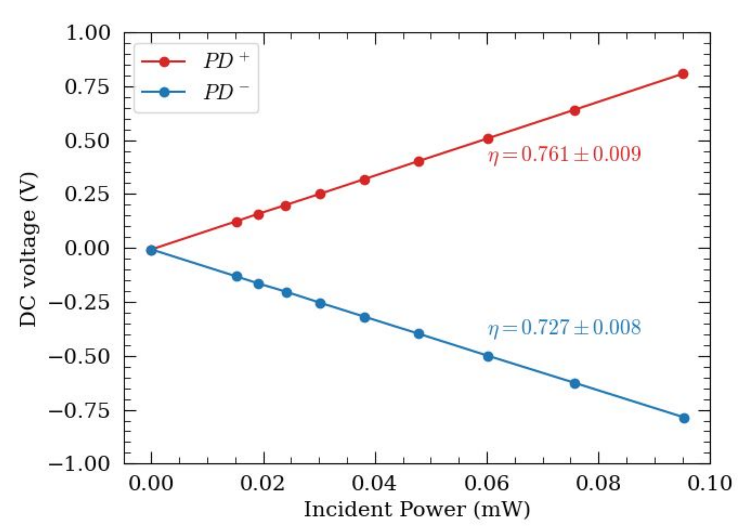

Fig. 5: Characterization of photodetectors.

The power spectral density of the HD was measured for different laser powers, to determine the optimal power range – avoiding both saturation and noise limit.

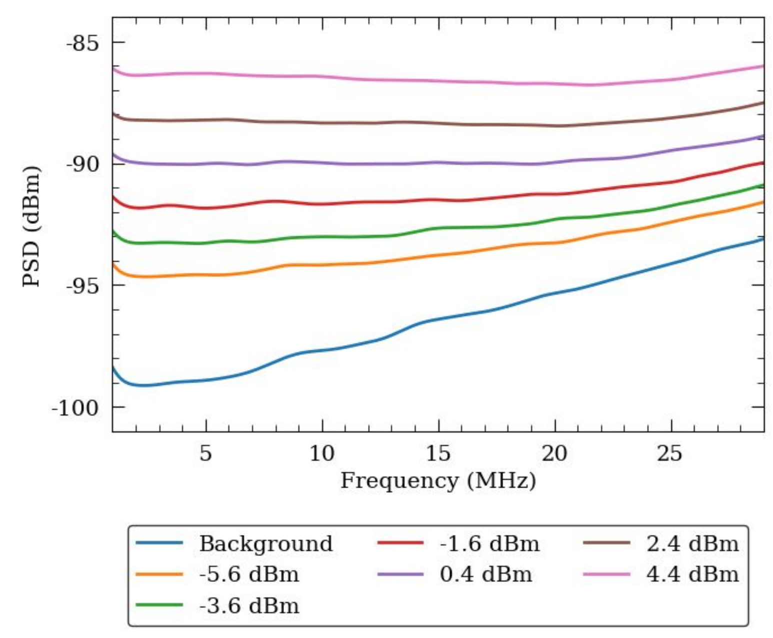

Fig. 4: Characterization of the Homodyne detector.

ii. Phase Shifter

The phase shifter is characterized by measuring the HD DC voltage with different voltages applied to the phase shifter. Using a ramp function we can scan the interference pattern and estimate the voltage needed for a π phase shift.

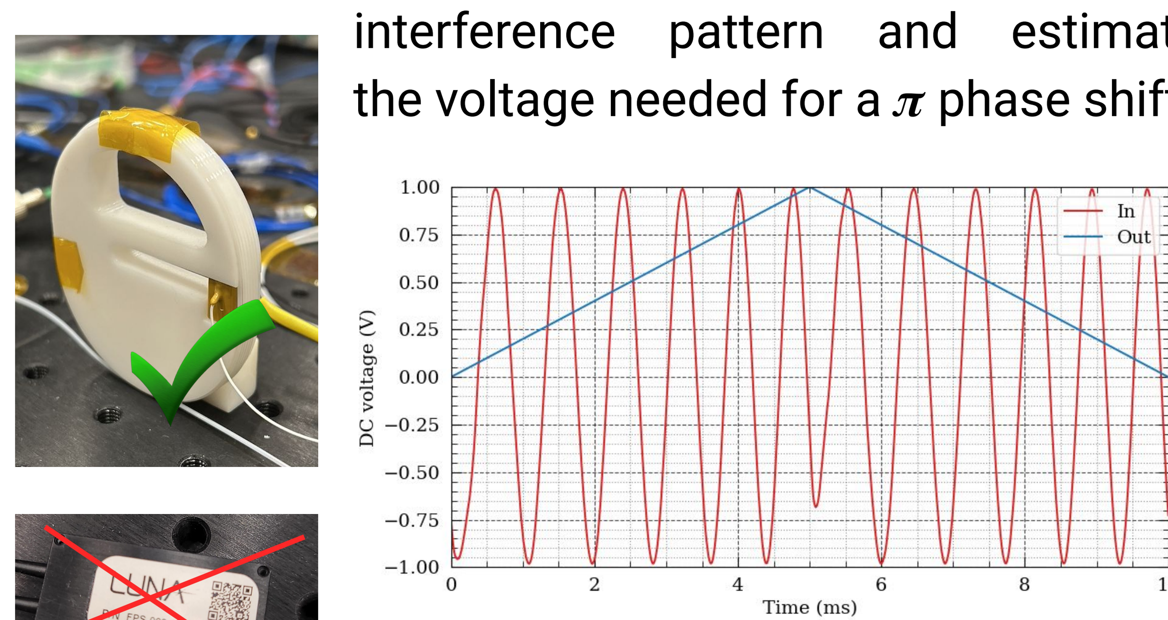

Fig. 6: (a)final p-s. (b)initial p-s. (c)resulting phase shift w.r.t. to voltage applied to piezo.

5. Implementation

Due to environmental noise, it was necessary to use the PID (Proportional-Integral-Derivative) controller module integrated in the Red Pitaya to lock the phase at DC = 0, and perform phase and amplitude modulation. Afterwards we could create the required signal from a python script, in order to see the given modulation.

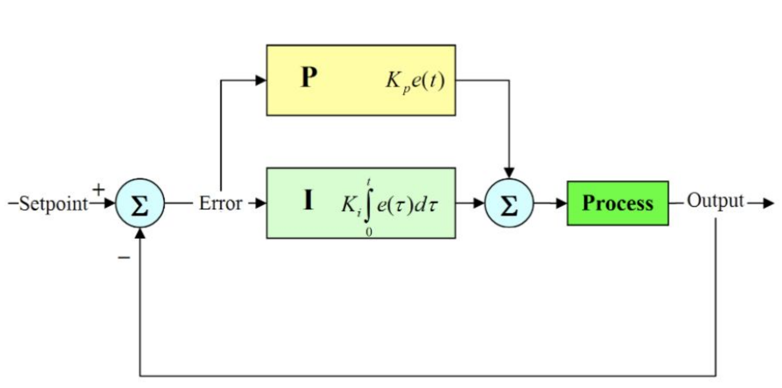

Fig. 7: Scheme for the PI(D) controller

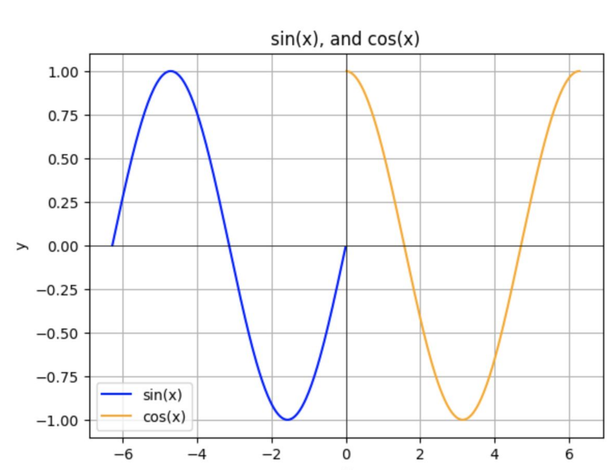

Fig. 8: Phase modulation achieved through shifting,

6. Results

We managed to create a signal from one Red Pitaya (Alice) oscillating between cosine and sine to another Red Pitaya (Bob). However, due to the time delays between the Red Pitayas, the modulation didn't work as the phases between the two weren't synchronized.

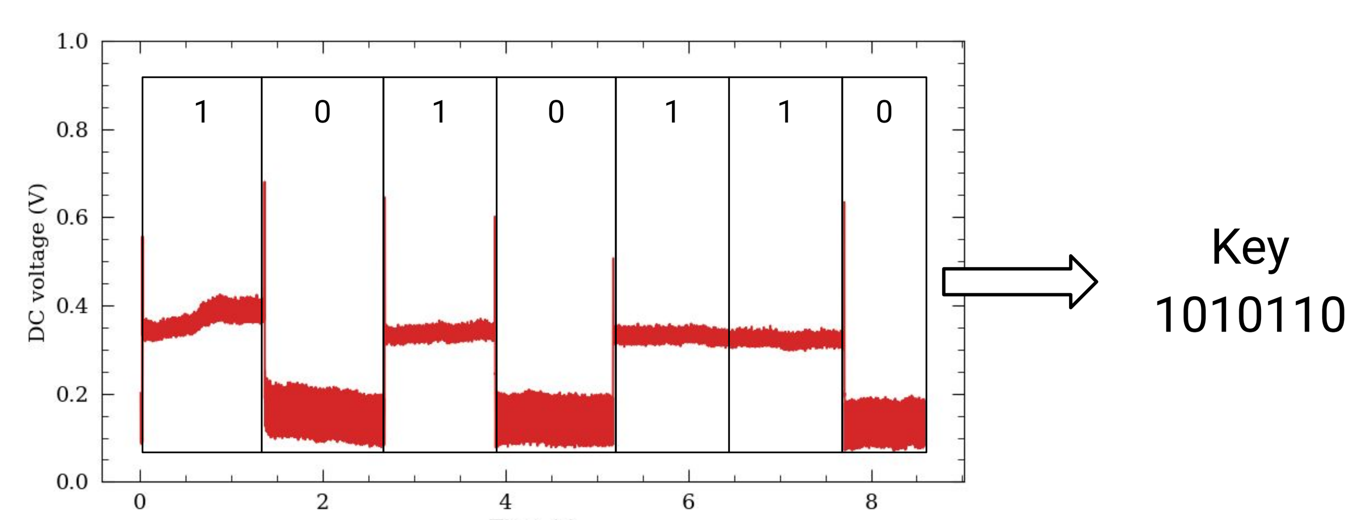

Fig. 9: AC signal measured after applying different phase modulations